

What is Data Cleaning?

Data Cleaning is the process of preparing raw data by correcting or removing inaccurate, incomplete, corrupted, or irrelevant data. It is a crucial step in the data preparation phase to ensure high-quality and reliable data, leading to more accurate analysis and insights.

In **Power Query**, a powerful data transformation tool integrated with Power BI, Excel, and other Microsoft products, data cleaning is done by applying various techniques and functions. Power Query automates much of the manual work involved in data cleaning through its user-friendly interface.

Techniques for Data Cleaning in Power Query

1. Handling Missing or Null Values:

- **Replace nulls with default values:** You can replace null values with default values such as 0, average, or the most common value.
 - **Power Query Function:** `Replace Values`
- **Remove rows with null values:** You can filter out rows that have missing or null values to avoid inaccurate calculations.
 - **Power Query Function:** `Remove Rows > Remove Blank Rows`

Example: Replacing all null values in the "Salary" column with 0.

Power

```
Table.ReplaceValue("#PreviousStep", null, 0, Replacer.ReplaceValue, {"Salary"})
```

2. Removing Duplicates

- Duplicate records can skew analysis. Power Query can help remove duplicate rows based on one or multiple columns.
 - **Power Query Function:** `Remove Duplicates`

Example: Removing duplicate rows based on the "CustomerID" column.

Power

```
Table.Distinct("#PreviousStep", {"CustomerID"})
```

3. Trimming and Cleaning Text:

- **Trim** removes extra spaces before and after text, while **Clean** removes non-printable characters.
 - **Power Query Function:** `Text.Trim, Text.Clean`

Example: Cleaning and trimming text in the "CustomerName" column.

Power

```
Table.TransformColumns("#PreviousStep", {"CustomerName", Text.Trim})
```

4. Splitting Columns:

- Sometimes, a single column contains multiple pieces of information that need to be split into different columns.
 - **Power Query Function:** `Split Column`
 - Methods: By delimiter (e.g., comma, space), by number of characters, by position.

Example: Splitting a "FullName" column into "FirstName" and "LastName" using a space delimiter.

Power

```
Table.SplitColumn("#PreviousStep", "FullName",  
Splitter.SplitTextByDelimiter(" "), {"FirstName", "LastName"})
```

5. Unpivoting and Pivoting Data:

- **Unpivot:** Converts columns into rows, which is useful when data is stored in a cross-tabulated format.
- **Pivot:** Converts rows into columns, ideal for summarizing data.

Example: Unpivoting columns containing monthly sales data into rows.

Power

```
Table.UnpivotOtherColumns("#PreviousStep", {"CustomerID"}, "Month",  
"Sales")
```

6. Replacing Values:

- You can replace specific values across a dataset, such as changing typos or standardizing text formats.
 - **Power Query Function:** `Replace Values`

Example: Replacing all occurrences of "USA" with "United States" in the "Country" column.

Power

```
Table.ReplaceValue("#PreviousStep", "USA", "United States",  
Replacer.ReplaceText, {"Country"})
```

7. Filtering Rows:

- Remove rows that don't meet certain criteria, such as removing rows with negative values or outliers.
 - **Power Query Function:** `Filter Rows`

Example: Removing rows where "Sales" is less than 0.

Power

```
Table.SelectRows("#PreviousStep", each [Sales] >= 0)
```

8. Merging Data:

- Combine multiple datasets into one by merging tables based on common columns (keys). This is used to clean and consolidate data from different sources.
 - **Power Query Function:** Merge Queries

Example: Merging "Sales" data with "Customer" data based on the "CustomerID".

Power

```
Table.NestedJoin("#SalesData", {"CustomerID"}, "#CustomerData",  
{"CustomerID"}, "MergedData", JoinKind.Inner)
```

9. Removing Errors:

- Power Query allows you to remove rows with errors that may arise during transformations (e.g., type conversion errors).
 - **Power Query Function:** Remove Errors

Example: Removing rows that contain errors in any column.

Power

```
Table.RemoveRowsWithErrors("#PreviousStep", {"Column1", "Column2",  
"Column3"})
```

10. Changing Data Types:

- Ensuring the correct data type is essential for accurate analysis. You can convert columns to appropriate types such as numbers, dates, or text.
 - **Power Query Function:** Change Data Type

Example: Converting the "SalesDate" column to Date type.

Power

```
Table.TransformColumnTypes("#PreviousStep", {"SalesDate", type date})
```

Common Power Query Functions for Data Cleaning

1. **Table.ReplaceValue:** Replaces values in a table based on conditions.
2. **Table.RemoveRowsWithErrors:** Removes rows that contain errors.
3. **Table.Distinct:** Removes duplicate rows from a table.
4. **Table.RemoveRows:** Removes rows that meet certain conditions.

5. **Table.TransformColumnTypes:** Changes the data type of columns.
 6. **Text.Trim, Text.Clean:** Cleans and trims text data.
 7. **Table.UnpivotOtherColumns:** Converts columns into rows for normalization.
 8. **Table.Pivot:** Converts rows into columns for summarization.
 9. **Table.SplitColumn:** Splits a column based on a delimiter or character count.
 10. **Table.Merge:** Merges two tables based on common columns (keys).
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Example Data Cleaning Workflow in Power Query

Let's say you have a dataset with customer information, but it contains issues like missing values, inconsistent country names, and extra spaces in customer names. Here's a step-by-step cleaning workflow using Power Query:

1. **Load Data:** Load your dataset into Power Query.
2. **Remove Duplicates:** Remove duplicate customer entries.

Power

```
Table.Distinct(#"PreviousStep", {"CustomerID"})
```

3. **Replace Nulls:** Replace any null values in the "Age" column with the average age.

Power

```
Table.ReplaceValue(#"PreviousStep", null, 35, Replacer.ReplaceValue, {"Age"})
```

4. **Trim Text:** Trim spaces in the "CustomerName" column.

Power

```
Table.TransformColumns(#"PreviousStep", {"CustomerName", Text.Trim})
```

5. **Standardize Country Names:** Replace inconsistent country names (e.g., "USA" with "United States").

Power

```
Table.ReplaceValue(#"PreviousStep", "USA", "United States", Replacer.ReplaceText, {"Country"})
```

6. **Change Data Types:** Convert "DateOfBirth" to a proper date format.

Power

```
Table.TransformColumnTypes(#"PreviousStep", {"DateOfBirth", type date})
```

After applying these transformations, you can load the cleaned data back into Power BI or Excel for further analysis.

Why Data Cleaning is Important

- **Improves Data Quality:** Clean data is accurate and complete, leading to better insights and decision-making.
- **Reduces Errors:** Eliminating duplicates, nulls, and inconsistencies reduces the risk of errors in analysis.
- **Saves Time:** Automated data cleaning using Power Query allows you to clean large datasets quickly.
- **Increases Usability:** Clean and well-organized data can be reused and shared easily across teams and applications.

By using Power Query's robust data transformation functions, you can automate most of the data cleaning tasks, making the process faster and less prone to human error.